Republic of Iraq
Ministry of Higher Education and Scientific Research
Computer Engineering
Computer Information Engineering

Power Flow Analysis in Electrical Grids Using Matrix Inversion Techniques

Prepared by:

حسين وسام جاسم محمد
رفل رياض حسن نصيف
زهراء علاء محمد باقر
زينب وليد جاسم رحيمه
سجاد احمد هاشم قاسم
سما محمد شاكر محمد

Stage: Second

Group: GA

Study type: Morning

Supervisor: Azhar Malik

Introduction

Electrical grids are among the largest systems ever built. They connect power plants to homes, factories and cities through many buses (nodes) and transmission lines. To run a grid safely, an engineer must know the voltage at every bus, the power flowing in every line, and how much power is lost on the way.

Power flow analysis (also called load flow) answers these questions. It is the base of most other grid studies, such as planning, stability and fault analysis. The grid is described by algebraic equations, so solving it means working with matrices. This is why **matrix inversion** and its related tools (LU decomposition, sparse matrices) matter so much: they decide how fast and how accurately the solution is found.

This report explains what power flow is, how the grid is modeled with the admittance matrix, and how matrix inversion is used to solve it. Simple numerical examples follow. In all examples, the matrix being worked on is called A.

1. What Is Power Flow Analysis?

Power flow analysis calculates the **steady-state** condition of a grid for known generation and load. It is a snapshot of the grid at one moment. Its main outputs are:

- the voltage magnitude and angle at every bus;
- the active power (P) and reactive power (Q) in every line;
- the power each generator must produce;
- the total losses (generated power minus consumed power).

The results are used for planning new plants and lines, checking that voltages stay within about ±5% of nominal, testing the loss of one line (the N-1 test), and reducing losses and fuel cost.

The problem is **nonlinear**, because power equals voltage times current and both are unknown. So it cannot be solved in one step. Iterative methods are used, and every iteration solves a linear system, which is where matrix inversion enters.

2. Modeling the Grid: The Admittance Matrix

Each line has a series impedance $z = r + jx$. Its inverse is the admittance:

$$y = 1/z = 1/(r + jx) = g + jb$$

Applying Kirchhoff's current law at every bus gives the matrix equation $\mathbf{I = Y V}$, where I is the vector of injected currents, V the bus voltages, and Y the $n \times n$ admittance matrix (Y-bus). Building Y is easy:

- diagonal element Y_{ii} = sum of all admittances connected to bus i;
- off-diagonal element Y_{ij} = minus the admittance of the line between buses i and j (zero if there is no line).

Y is symmetric and **sparse**: most of its entries are zero. Its inverse $Z = Y^{-1}$ is the impedance matrix, and it is dense. This loss of sparsity is the main reason large grids avoid computing Z directly.

Example 1: Bus voltages of a three-bus grid

A grid has line conductances $g_{12} = 2$, $g_{13} = 1$ and $g_{23} = 1$ (per unit). Using the rules of Section 2:

$$Y = \begin{vmatrix} 3 & -2 & -1 \\ -2 & 3 & -1 \\ -1 & -1 & 2 \end{vmatrix}$$

Every row sums to zero, so Y is singular (Example 4). Take bus 1 as slack with $V_1 = 1$ and remove its row and column. The reduced matrix is A:

$$A = \begin{vmatrix} 3 & -1 \\ -1 & 2 \end{vmatrix} \Rightarrow A^{-1} = 1/5 \times \begin{vmatrix} 2 & 1 \\ 1 & 3 \end{vmatrix}$$

The loads draw 0.1 at bus 2 and 0.2 at bus 3 ($I_2 = -0.1$, $I_3 = -0.2$). Then $V = V_1 + A^{-1} I$ gives $V_2 = 1 + (2(-0.1) + 1(-0.2))/5 =$ **0.92** and $V_3 = 1 + (1(-0.1) + 3(-0.2))/5 =$ **0.86**. Check at bus 3: $-1 - 0.92 + 2(0.86) = -0.2 = I_3$. ✓

3. Bus Types

Each bus has four quantities: voltage |V|, angle θ, active power P and reactive power Q. Two are known and two are unknown, which gives three bus types:

Bus type	Known	Unknown	Example
Load bus (PQ)	P, Q	|V|, θ	A city or factory
Generator bus (PV)	P, |V|	Q, θ	A controlled generator
Slack bus	|V|, θ	P, Q	Main generator

The slack bus is needed because losses are not known in advance, so one generator must cover them. Its angle is set to zero and is the reference for all other angles. The power equations contain sine and cosine terms, so they are nonlinear.

4. The Role of Matrix Inversion

Nearly every power flow method ends with a linear system **A x = b**, whose solution is $x = A^{-1} b$. The inverse is used to find voltages from currents ($V = Z I$), to correct the guess in Newton-Raphson (the Jacobian inverse), and to study how a change at one bus affects the others. A matrix has an inverse only if its determinant is not zero. For a 2 × 2 matrix:

$$A = \begin{vmatrix} a & b \\ c & d \end{vmatrix} \Rightarrow A^{-1} = 1/(ad - bc) \times \begin{vmatrix} d & -b \\ -c & a \end{vmatrix}$$

For large grids direct inversion is too slow (about n^3 operations), so programs use better techniques:

- **LU decomposition:** split A = L U, then solve by two quick steps (forward and backward substitution). A^{-1} is never formed.
- **Sparse techniques:** store only non-zero entries and reorder the buses to reduce extra non-zeros.
- **Gauss-Seidel:** needs no inversion but is slow on large grids.

Example 2: Inverse of a 2 × 2 matrix

Find A^{-1} for the matrix below.

$$A = \begin{vmatrix} 2 & 1 \\ 1 & 1 \end{vmatrix}$$

Determinant $= (2)(1) - (1)(1) = 1$, not zero, so the inverse exists. Swap the diagonal, change the signs of the other two entries, and divide by the determinant:

$$A^{-1} = \begin{vmatrix} 1 & -1 \\ -1 & 2 \end{vmatrix}$$

Example 3: Solving A x = b with the inverse

Using the A of Example 2, solve $A\,x = b$ with $b = [5, 3]^T$. This means $2x_1 + x_2 = 5$ and $x_1 + x_2 = 3$.

$$x = A^{-1}\,b = \begin{vmatrix} 1 & -1 \\ -1 & 2 \end{vmatrix} \begin{vmatrix} 5 \\ 3 \end{vmatrix} = \begin{vmatrix} 2 \\ 1 \end{vmatrix}$$

So $x_1 = 2$ and $x_2 = 1$. Check: $2(2) + 1 = 5$ and $2 + 1 = 3$. ✓

Example 4: A matrix that cannot be inverted

$$A = \begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix}$$

The determinant is $(1)(4) - (2)(2) = 0$, so A is **singular**: the second row is just twice the first. This matters for grids: the full Y-bus is always singular because every row sums to zero. That is why one bus (the slack bus) is removed before inverting.

Example 5: Solving with LU decomposition

Solve $A\,x = b$ with $b = [3, 9]^T$, without computing A^{-1}. First split A into L and U:

$$A = \begin{vmatrix} 2 & 1 \\ 4 & 5 \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 2 & 1 \end{vmatrix} \begin{vmatrix} 2 & 1 \\ 0 & 3 \end{vmatrix} = L\,U$$

Forward step ($L\,y = b$): $y_1 = 3$, then $2(3) + y_2 = 9$, so $y_2 = 3$. **Backward step** ($U\,x = y$): $3x_2 = 3$, so $x_2 = 1$; then $2x_1 + 1 = 3$, so $x_1 = 1$. The solution is $x = [1, 1]^T$. If b changes, only these two quick steps are repeated.

6. Newton-Raphson in Brief

Newton-Raphson is the most used power flow method, usually converging in 3 to 5 iterations. In each iteration the power mismatches ΔP and ΔQ are turned into corrections by inverting the Jacobian matrix, which we call A:

$$[\Delta\theta,\ \Delta|V|]^T = A^{-1}\ [\Delta P,\ \Delta Q]^T$$

Because A is inverted in every iteration, this step controls the speed of the whole program.

Example 6: A simplified (DC) power flow

The DC power flow uses only angles. Take susceptances $b_{12} = 10$, $b_{13} = 5$, $b_{23} = 5$, bus 1 as slack ($\theta_1 = 0$), and loads of 0.8 and 0.2 pu at buses 2 and 3. Then $A\,\theta = P$ with:

$$A = \begin{vmatrix} 15 & -5 \\ -5 & 10 \end{vmatrix}, \quad P = \begin{vmatrix} -0.8 \\ -0.2 \end{vmatrix}$$

The determinant is $15(10) - 25 = 125$, so:

$$\theta = A^{-1}\,P = 1/125 \times \begin{vmatrix} 10 & 5 \\ 5 & 15 \end{vmatrix} \begin{vmatrix} -0.8 \\ -0.2 \end{vmatrix}$$

$$= \begin{vmatrix} -0.072 \\ -0.056 \end{vmatrix}$$

Line flows $P_{ij} = b_{ij}(\theta_i - \theta_j)$: $P_{12} = 0.72$, $P_{13} = 0.28$, $P_{23} = -0.08$. Check: the slack supplies 0.72 + 0.28 = 1.0 pu, equal to the total load.

7. Gauss-Seidel Method

Gauss-Seidel is the simplest way to solve **A x = b** without inverting anything. It starts from a guess and improves it round after round, until the answer stops changing.

The steps are simple:

- Solve each equation for its own unknown.
- Start from a guess (usually all zeros).
- Update the unknowns one by one, always using the newest values.
- Repeat until two rounds give almost the same numbers.

It needs very little memory and works well when the diagonal entries of A are the largest in their rows, as in grid matrices. Its weakness is speed: large grids need many rounds, so Newton-Raphson is preferred in practice.

Example 7: Gauss-Seidel on the three-bus grid

Use the reduced matrix A of Example 1 and the same loads, so A x = I with I = $[-0.1, -0.2]^T$. Here x_1 and x_2 are the voltage changes at buses 2 and 3. The equations are $3x_1 - x_2 = -0.1$ and $-x_1 + 2x_2 = -0.2$. Solve each one for its own unknown:

$$x_1 = (-0.1 + x_2) / 3$$
$$x_2 = (-0.2 + x_1) / 2$$

Start with $x_1 = x_2 = 0$ and repeat. Each new x_1 is used at once to find x_2:

Round	x_1	x_2
1	−0.0333	−0.1167
2	−0.0722	−0.1361
3	−0.0787	−0.1394
4	−0.0798	−0.1399
Exact (Example 1)	−0.0800	−0.1400

After 4 rounds $V_2 = 1 + x_1 = 0.920$ and $V_3 = 1 + x_2 = 0.860$, the same as in Example 1. The inverse gave the answer in one step; Gauss-Seidel needed four rounds but never formed A^{-1}.

Conclusion

Power flow analysis is the basic tool for understanding, planning and operating electrical grids. The grid is modeled with the admittance matrix, and the nonlinear power equations are solved step by step. Matrix inversion appears at every stage. Since direct inversion is expensive for large grids, real software uses LU decomposition, sparse storage and fast decoupled methods. The examples showed that the same idea (build A, invert or factorize it, extract the unknowns) works from a 2 × 2 toy problem up to thousands of buses.

References

[1] Glover, J. D., Overbye, T. J., & Sarma, M. S. *Power System Analysis and Design*. Cengage Learning.

[2] Saadat, H. *Power System Analysis*. McGraw-Hill.

[3] Grainger, J. J., & Stevenson, W. D. *Power System Analysis*. McGraw-Hill, 1994.

[4] Kundur, P. *Power System Stability and Control*. McGraw-Hill, 1994.

[5] Tinney, W. F., & Hart, C. E. "Power Flow Solution by Newton's Method." *IEEE Trans. Power Apparatus and Systems*, vol. PAS-86, no. 11, pp. 1449–1460, 1967.

[6] Stott, B., & Alsac, O. "Fast Decoupled Load Flow." *IEEE Trans. Power Apparatus and Systems*, vol. PAS-93, no. 3, pp. 859–869, 1974.

[7] Stott, B. "Review of Load-Flow Calculation Methods." *Proceedings of the IEEE*, vol. 62, no. 7, pp. 916–929, 1974.